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ORIGINAL RESEARCH REPORT

Pulsed Red and Blue Photobiomodulation for the Treatment of Thigh Contusions and Soft Tissue Injury: A Randomized Controlled Trial

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Context: Contusion and soft tissue injuries are common in sports. Photobiomodultion, light and laser therapy, is an effective aid to increase healing rates and improve function after various injury mechanisms. However, it is unclear how well photobiomodulation improves function after a contusion soft tissue injury. This study aimed to determine the effects of a pulsed red and blue photobiomodulation light patch on muscle function following a human thigh contusion injury. ***Design:*** Single-blinded randomized control trial design. ***Methods:*** We enrolled 46 healthy participants. Participants completed 5 visits on consecutive days. On the first visit, participants completed a baseline isokinetic quadriceps strength testing protocol at 60°/s and 180°/s. On the second visit, participants were struck in the rectus femoris of the anterior thigh with a tennis ball from a serving machine. Immediately following, participants were treated for 30 minutes with an active or placebo photobiomodulation patch (CareWear light patch system, CareWear Corp). Following the treatment, participants completed the same isokinetic quadriceps strength testing protocol. Participants completed the treatment and isokinetic quadriceps strength test during the following daily visits. We normalized the data by calculating the percent change from baseline. We used a mixed model analysis of covariance, with sex as a covariate, to determine the difference between treatment groups throughout the acute recovery process. ***Results:*** We found the active photobiomodulation treatment significantly increased over the placebo group, quadriceps peak torque during the 180°/s test ($P = .030$), and average power during both the 60°/s ($P = .041$) and 180°/s ($P \leq .001$) assessments. The mean peak torque and average power of 180°/s, at day 4, exceeded the baseline levels by 8.9% and 16.8%, respectively. ***Conclusions:*** The red and blue photobiomodulation light patch improved muscle strength and power during the acute healing phase of a human thigh contusion injury model.

Keywords: light therapy, bruise, recovery

Contusions and soft tissue injuries are common in sport, making up approximately 12.1% of all injuries,[1] and account for 8% to 22% of all time-loss injuries in various high school and college sports.[2–6] Researchers have reported the average return to activity without proper treatment of a thigh contusion at 45 days; however, a mild thigh contusion with appropriate treatment returned to activity on average in 6.5 days.[7] For a quick return to activity for patients, it is imperative that clinicians apply the proper treatment that limits the size of the hematoma, repairs the damaged muscle and capillary tissue, and improves function after a contusion and soft tissue injury. Large hematoma formations will decrease knee range of motion, increase quadriceps muscle stiffness, and reduce muscle strength.[8,9] Additional bleeding, edema, and ischemia are potential barriers to healing and may cause increased scar formation.

The typical acute treatment for contusions and soft tissue injuries is cryotherapy and nonsteroidal anti-inflammatory drugs. Immediate care cryotherapy may lower the tissue metabolic rate protecting from further secondary injury.[10] However, the acute clinical effectiveness of cryotherapy is still unclear for musculoskeletal injuries.[11,12] Nonsteroidal anti-inflammatory drugs may aid pain management, but mixed theories exist about their effects on the healing process following muscle injury.[13]

Photobiomodulation (PBM) has had positive outcomes in healing tissue and improving function. Animal contusion models indicate PBM effectively reduces oxidative stress markers after muscle injury.[14] PBM treatments in human models after eccentric exercise bouts have found increased healing rates and improved muscle function.[15,16] PBM's primary mechanism of stimulating an increase in mitochondrial adenosine triphosphate (ATP) production and dissociation of nitric oxide may positively create an acute tissue-protective effect and fuel earlier tissue healing during the injury sequelae of a contusion.

A new type of PBM light patch has been developed and found to have positive clinical outcomes with reducing muscle fatigue.[17] The light patch delivers PBM pulsed red and blue wavelength energy evenly over the treatment area. Currently, there is limited evidence on the effects of this red, 640 nm, and blue, 450 nm, wavelength combination for muscle recovery after injury, as the majority of literature focuses on the red and infrared wavelength spectrum. However, the light patch allows for ease of use within the clinical setting enabling the patient to move unobstructed and freeing up time for the clinician. Understanding the clinical outcomes of this new light patch technology is essential to providing appropriate and proper treatment for debilitating contusion and soft tissue injuries. The purpose of this study was to determine the effects of a new PBM light patch, with pulsed red and blue wavelengths, on muscle isokinetic strength and power during a human thigh contusion injury, using an experimental injury model.

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Materials and Methods

Study Design

We used a single-blinded randomized control trial study design for this study. The study's independent variables included the treatment group (active vs placebo PBM) and 5 different time points. We blinded the participants to their treatment group allocation. The dependent variable of this study was isokinetic quadriceps strength at 180°/s and 60°/s. This main outcome measure is reliable and valid,[18,19] and other researchers have used this primary outcome in assessing muscle performance after various therapeutic interventions, including PBM.[15,16,20]

Participants

We enrolled 46 healthy participants (male = 23, female = 23, age = 21.9 [1.7] y, mass = 80.4 [16.6] kg) for this clinical trial. Participants met the study's inclusion and exclusion criteria if they had no prior or current lower-extremity injury within the past 6 weeks, had no history of blood clotting or bruising complications, and were generally healthy. Before enrolling participants in the research study, the institutional review board approved the study protocol at Brigham Young University, and all participants provided informed consent.

An a priori sample size estimate was determined using power software (G*Power, version 3.1.9.4; Kiel University). A sample size of 22 participants per group was determined necessary using an alpha of .05, a beta of 0.20, and a between-group (PBM and placebo) percent change effect size of $d = 0.35$.

Light Patch Application

We use the CareWear light patch system (CareWear Corp). This PBM system uses a 50 cm^2 wearable light patch containing 130 hexagon-shaped clusters of micro-light-emitting diodes of blue and red wavelengths (450 and 640 nm). The light patch is powered by a small wireless controller magnetically connected. The ratio of blue to red optical power is 3:1. The light patch system has an average irradiance of 2.5 mW/cm^2 and operates in a high-frequency pulsed mode of 33 kHz, at a 33% duty factor, for a peak irradiance of 7.5 mW/cm^2. We set the treatment duration for 30 minutes, creating a total energy delivery of 225 J at a fluence of 4.5 J/cm^2. These parameters match established dosage ranges for PBM's effectiveness in tissue repair,[21] pain management,[22] and improved muscle performance and recovery.[17]

Procedures

Participants made 5 total visits to the research laboratory (Figure 1). Participants completed the informed consent process at their first visit, and the same investigator (AW) screened them for the inclusion and exclusion criteria using a self-reported health questionnaire. Participants then completed baseline quadriceps strength and power tests using an isokinetic dynamometer (Biodex system 4-Pro, Biodex Medical Systems, Inc). We used the same repetitions and sets for both isokinetic speeds tested to simplify the instructions and ensure participant consistency across tests. Participants performed the isokinetic test using their dominant leg, consisting of 5 repetitions of knee flexion and extension at a speed of 180°/s, followed by 30 seconds of rest and another set of 5 repetitions for a total of 10 repetitions.[23,24] The participants then performed the same isokinetic test routine (5 reps × 2 set with 30-s rest in between) at 60°/s. The same investigator then allocated participants into groups using a stratified randomization method to ensure equal sex distribution between groups. Participants randomly drew a number out of a bag to assigned into one of 2 groups: (1) active pulsed red and blue light treatment or (2) placebo (N = 23 in each group). We used different bags filled with randomization numbers for males and females.

On the second visit, we bruised the participants' quadriceps by using a tennis ball serving machine. This human thigh contusion injury model has been reported previously in the literature.[25,26] The participant was placed in front of a tennis ball serving machine at a distance of 30 cm, with bodily protection leaving only the anterior thigh exposed. The investigator visualized the midpoint of the rectus femoris and lined the participants up to receive a direct inline shot. During the protocol development, we checked these procedures using imaging ultrasound (GE model logic S8, 6–15 MHz linear probe, 11 MHz frequency) to confirm the location of the ball contact to the rectus femoris. We were able to visualize immediate edema formation and rectus femoris muscle damage up to 2 cm deep. Participants were struck with a new tennis ball by the machine set at a 16-speed setting (approximately 112 kph). The investigator visually ensured that each participant received the same ball force and anatomical location to create the contusion injury. Within 10 minutes of the injury, participants received a 30-minute active or placebo PBM treatment while lying supine (Figure 2), depending on their randomly assigned treatment group. We placed a towel between the light patch and the participants' vision to prohibit participants from knowing whether they were in the PBM treatment group or placebo group. Immediately after the treatment, participants performed the same 180°/s and 60°/s isokinetic quadriceps testing protocol (5 repetitions × 2 sets with 30-s rest period between sets). We instructed participants not to apply any treatment to the area or take any medications during the duration of the study.

Participants reported to the laboratory 3 additional consecutive days for visits 3 to 5. Each session began with a 30-minute treatment session, followed by the isokinetic quadriceps testing protocol. After the fifth visit, we excused the participants from the study.

Statistical Analysis

We initially normalized the data by calculating the percent change from baseline measurement for all the isokinetic outcome measures, peak torque, and average power at 180°/s and 60°/s. We calculated descriptive statistics. To determine the difference between treatment groups throughout the acute muscle recovery phase, we used a mixed model analysis of covariance (ANCOVA), using sex as a covariate, with interest in the treatment condition × time interaction. For the 180°/s analysis, sex was used as a covariate, but sex was not a significant covariate in the 60°/s analysis, thus not used in this analysis. When appropriate, we used a Tukey–Kramer post hoc analysis. We completed all data analysis using JMP Pro (version 15, JMP Statistical Discovery LLC) and set the α level at .05.

Results

All participants started the treatment condition as randomly allocated. Two participants' data, one from each PBM group, were excluded from the statistical analysis due to the lack of study protocol compliance. Therefore, the study's participant completion rate was 95.7% (44/46, final N = 22 in each treatment group). The data were inspected and found to meet the assumptions from proper analysis using an ANCOVA.

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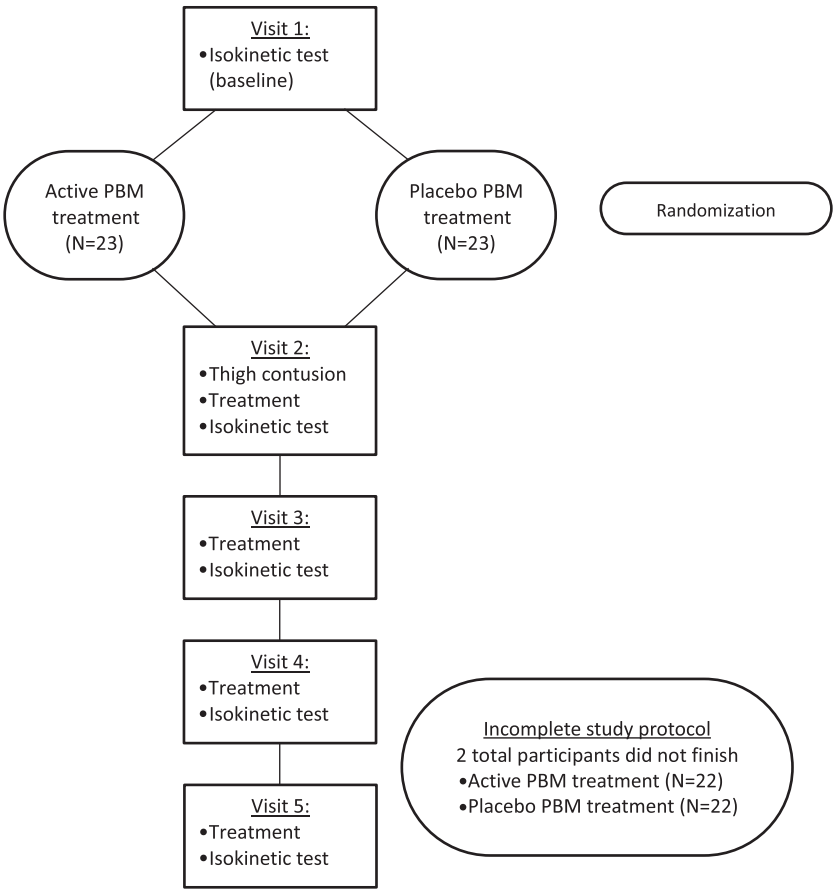

Figure 1 — Flow diagram of study procedures. The knee isokinetic test occurred at 60°/s and 180°/s. Participants obtained a thigh contusion on visit 2. Treatments were provided based on their randomly assigned treatment group. PBM indicates photobiomodulation.

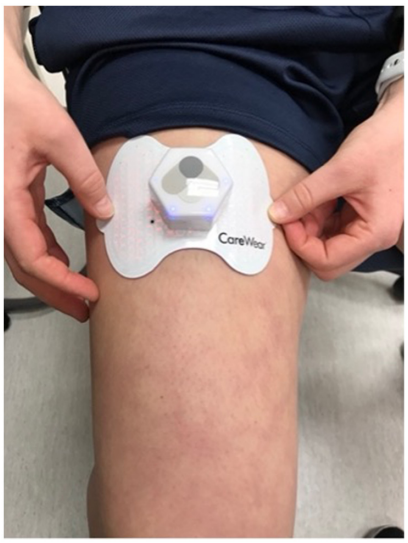

Figure 2 — Active photobiomodulation treatment over quadriceps muscle. A single patch was placed directly over the thigh contusion.

Isokinetic Speed 180°/s

The active treatment group had a significant improvement in quadriceps peak torque ($F_{4,167} = 2.74$, $P = .030$, sex covariate, $P = .013$) and average power ($F_{4,167} = 4.70$, $P = .001$, sex covariate, $P = .041$) over the 4-day treatment period (treatment condition × time interaction). At the end of the 4 treatment days, the active treatment group had increased their quadriceps 180°/s peak torque and average power by 8.9% (12.1%) and 16.8% (20.5%) above baseline, respectively. Conversely, the placebo treatment group did not return to above baseline isokinetic strength values at the end of the 4 treatment days (peak torque = −1.7% [15.5%], average power = −2.3% [20.8%]; Figure 3).

Isokinetic Speed 60°/s

At day 4, the peak torque percent change from baseline was −1.7% (14.2%) and −5.5% (11.5%) for the active and placebo groups, respectively. There was no difference between sex in either peak torque ($P = .181$) and average power ($P = .065$) at 60°/s. At this slower isokinetic speed, there was no difference in peak torque between treatment groups over the 4-day treatment period (treatment condition × time interaction; $F_{4,167} = 0.54$, $P = .704$). However, there was a greater improvement in average power in the active treatment group over the 4-day treatment period ($F_{4,167} = 2.55$, $P = .041$). Quadriceps 60°/s average power returned to baseline strength values in the active treatment group at the end of the 4 treatment days but not in the placebo treatment group (Figure 3). The average power percent change from baseline was 1.5% (15.6%) in the active treatment group and −8.0% (11.0%) in the placebo treatment group.

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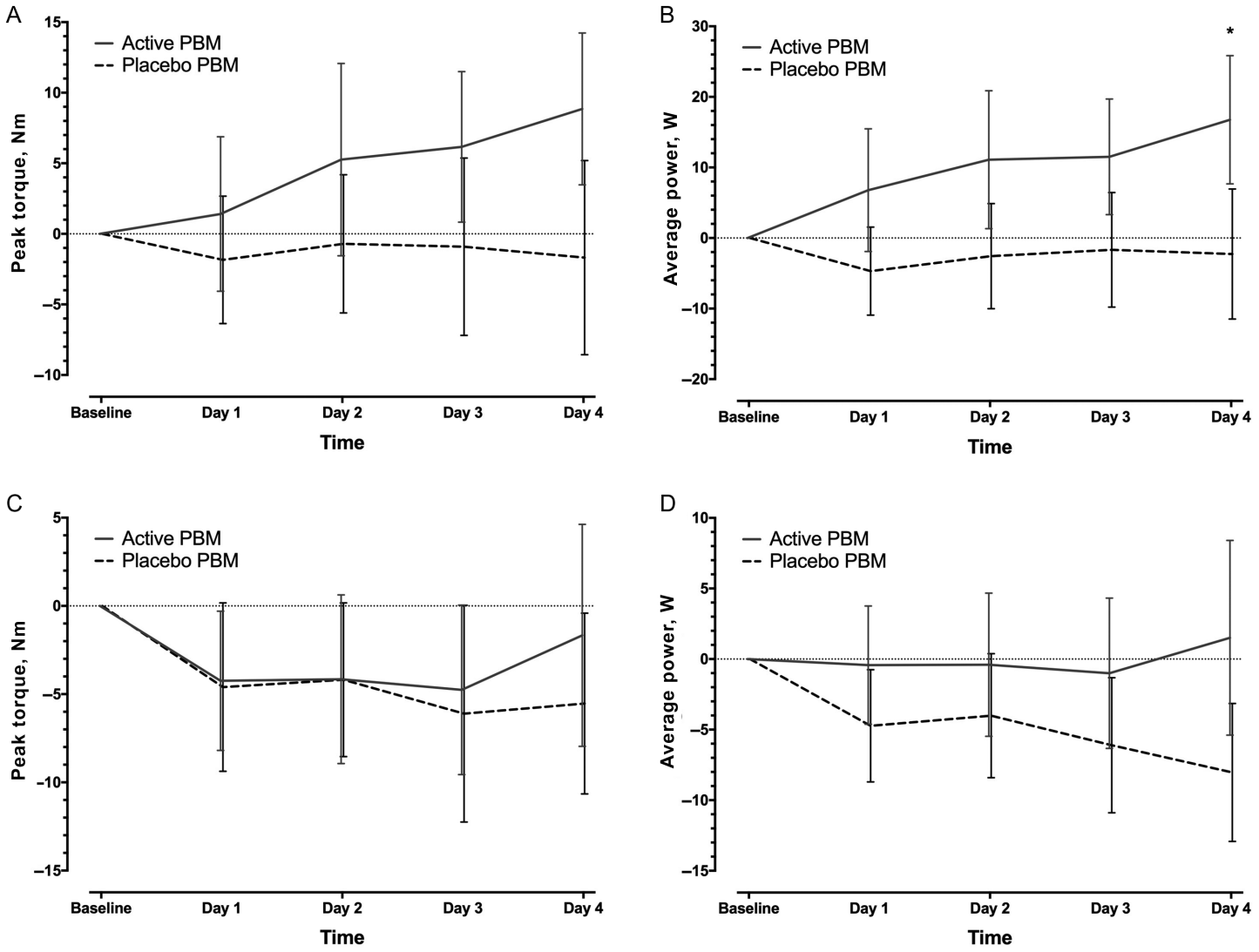

Figure 3 — Isokinetic strength value means with 95% confidence intervals at baseline and after 4 days of active or placebo PBM treatment. (A) 180°/s peak torque, (B) 180°/s average power, (C) 60°/s pak torque, and (D) 60°/s average power. *Active treatment had significant improvement of strength value over placebo treatment. PBM indicates photobiomodulation.

Discussion

Thigh contusion and soft tissue injuries caused by blunt, nonpenetrating objects account for a large portion of injuries in sport[1,2] and activities of daily living.[27] These soft tissue injuries lead to an inflammation and lowered healing response as the damage occurs to the muscles' soft tissue and vascular components surrounding the area of trauma. The healing response may lead to unnecessary fibrotic scar formation, limiting muscle force production, and may cause incomplete recovery of muscle strength. However, the improvement in isokinetic muscle strength and power we found during the acute contusion injury, especially at the isokinetic speed of 180°/s, indicates enhanced muscle strength produced by the pulsed red and blue PBM.

Improvements in isokinetic muscle strength after a contusion injury can be explained by the physiological effects of the used PBM wavelengths. In general, red wavelengths (620–660 nm) stimulate increased mitochondrial production of ATP through stimulation of cytochrome C oxidase, the final enzyme in the electron transport chain. The additional cellular energy provided by ATP is particularly important in accelerating cellular metabolism,[28] thereby accelerating fibroblastic production of collagen for tissue repair. Red and blue wavelengths (425–460 nm) stimulate the release of nitric oxide trapped in ischemic cells. When PBM stimulates nitric oxide dissociation, cellular respiration is improved and signaling increases vasodilation and oxygenation in the regional microcirculation.[29]

Contusion and soft tissue injuries often produce large hematomas, which may limit cellular respiration and cause secondary ischemic injury. The release of cytokines as a result of acute soft tissue injury is associated with inflammation, and peripheral sensitization, and is generally a short-lived reversible event.[30] Vascular and lymphatic clearance in the early stages of healing and modulation of proinflammatory cytokines is critical in reducing the likelihood of sustained injury.[31] Accelerating the inflammatory phase of healing is critical to allow for a quick return to normal activities and physical activity. PBM, especially with red and blue wavelengths, may combat this acute injury response and further speed up the tissue healing

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process by improving mitochondrial function and cellular respiration.[28,32] PBM's acceleration of the healing response causes a reduction of proinflammatory cytokines interleukin-1A, interleukin-1B and interleukin-6, tumor necrosis factor-alpha and is involved in cyclo oxygenase inhibition to decrease inflammatory cell proliferation.[14,33] Combined pulsed red and blue PBM light patch may increase acute tissue healing leading to greater muscle function and strength improvement seen in our results. Future research should measure tissue healing markers after combined red and blue PBM therapy.

In addition to aiding cellular respiration and the inflammatory response, PBM is also beneficial in stimulating stem cells and progenitor cells.[34] In vivo and in vitro animal injury models have shown an increase in satellite cell activation around myofibrils treated with PBM,[35,36] leading to a rise in new myofibril formation.[37,38] Irradiated muscle tissues promote a notable maturation of young myofibrils compared with those not irradiated.[39] Overall, muscle tissue treated with PBM after injury has a reduction of muscle atrophy.[40] Future research regarding muscle tissue healing after PBM needs to be conducted in human studies. However, it is encouraging that our results indicate enhanced muscle strength recovery after a contusion injury.

The PBM treatment had different effects on muscle performance at the 2 isokinetic speeds. The PBM treatment created physiological effects that altered improvements in peak torque at 180°/s, but not 60°/s. However, there was a significant improvement of average power at both isokinetic speeds. The force–velocity relationship may help explain these findings. Human skeletal muscle force decreases in a curvilinear fashion with increasing contractile velocities. In contrast, muscle power output increases to approximately one-third of maximum contractile velocity before exhibiting a parabolic decline.[41] Average power may be a better marker of muscle healing as it also accounts for the muscle to maintain its force production throughout the whole movement instead of a single instance of peak torque. The average power differences between treatment groups indicate that active PBM treatment created quicker quadriceps muscle strength recovery. However, full muscle strength recovery may not have occurred yet as indicated by difference in peak torque results at the 2 isokinetic speeds. The differences in peak torque at 60°/s indicate that additional muscle strength recovery is needed for the muscle to create high levels of torque and force at these slower movement speeds. There are positive acute muscle strength improvements with PBM over the placebo treatment, and additional research should determine a more prolonged time course of muscle strength and function using this treatment after an acute soft tissue injury.

We used a PBM light patch with a lower irradiance and uniform distribution of light over the treatment site than many PBM devices[20,42]; however, the light patch significantly improved muscle strength during the healing phase of an acute contusion. PBM irradiance and fluence may play a larger role in determining the PBM's tissue effects. The rate at which photons are applied to the tissue should match the cellular response.[28] There is an important biphasic dose–response relationship, and high irradiance treatments with high doses of energy may lead to different physical processes and diminished clinical outcomes.[43] Also, recent research indicates that the beam profile and surface area of irradiation significantly impact mitochondrial activity and thus ATP production. A uniform fluence distribution over the treatment area is important in maximizing this effect.[44] Our research indicates that using a flexible light patch which produces uniform distribution of light over the treatment site provides clinically significant muscle strength and power improvements during injury recovery at lower irradiances than standard handpieces at higher irradiances.

Typically, PBM has been applied to tissue using point stimulation, cluster probes, or light-diffusing optics. These require the clinician to hold or move a fiber optic, probe, or cluster applicator over the treatment area. Applying PBM in this manner is time-consuming and is not labor efficient, often leading to reduced dose and efficacy. The new PBM light patch we used uses printed light-emitting diodes on a flexible substrate. We applied the light patch directly to the skin over the area of contusion using a transparent hydrogel as an adhesive light pipe. Thus, the treatment was easy to apply and clinically efficient.[45]

This study may be limited by only assessing isokinetic muscle strength. However, using a reliable thigh contusion injury model is progress in understanding the benefits of PBM for soft tissue injury. PBM produces other strength-related positive healing effects; however, researchers have only observed this effect in exercise-induced muscle damage studies.[20,46] Our thigh contusion injury model produces a different mechanism versus an exercise-induced muscle damage model. However, additional research should continue to expand our understanding of this thigh contusion model. Currently, the injury model has only been assessed for healing using superficial photographic color analysis methods.[25] Additional imaging studies should be used to understand the extent of the injury. PBM produces positive effects in both injury models indicating important muscle strength recovery.[43] Future studies are needed to understand tissue oxygenation, vascular and lymphatic clearance, and changes in healing rates in our thigh contusion injury model.

An additional limitation of this study was that we did not measure pain associated with the contusion injury. There is a significant correlation between pain and isokinetic torque where pain could influence strength output.[47] Inversely, an improvement in strength is correlated with improvements in pain subscale outcomes on the 36 item short form pain questionnaire (SF-36), in recreation athletes with soft tissue injuries.[48] Beside strength, we are unable to quantify the extent of soft tissue injury that occurred, where additional measures such as imaging ultrasound or pain would have helped. We assumed that a similar injury occurred between participants using our consistent contusion injury mechanism. However, individual factors, such as subcutaneous fat thickness, could have altered the extent of the injury. We did stratify our allocation to ensure the same number of male and female were in each treatment group. Thus, we are only able to determine the effects of PBM on muscle strength during the acute recovery of the contusion injury.

Conclusions

Muscle contusion and soft tissue injuries are seen commonly in athletes and the average person performing various activities.[1] Often the effects of these injuries can result in significant time loss.[2–5] The use of PBM in treating muscle and soft tissue injury can be an excellent alternative to traditionally used forms of treatment. The active PBM pulsed red and blue light patch significantly improved muscle strength and power during an acute injury recovery as compared to the placebo treatment. Clinicians should consider the use of pulsed red and blue PBM following musculoskeletal trauma as a standard of care.

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